

Freight Rate Prediction — Validation Approach & December Forecast

Machine Learning Engineer Assessment · Model: CatBoost (selected over XGBoost)

1. Train / Test Validation Approach

Split methodology

The labeled development data (train-test.csv, 48,000 rows) was split into an 80% training set and a 20% held-out test set using a random split (scikit-learn `train_test_split`, `random_state=42` for reproducibility). A random split was chosen over a time-based split because the goal is to validate general predictive accuracy across the full range of routes, equipment types, and market conditions represented in the data, rather than to simulate strict forward-in-time forecasting. The held-out 20% (9,600 rows) is never seen during training and is used exclusively to compare candidate models and report final metrics.

Data-quality checks performed before splitting

- **Negative weight values** (292 rows, ~0.6%): physically impossible; treated as a sign error and corrected with an absolute-value transform rather than dropped, since the magnitude of the values looked plausible once corrected.
- **Missing values** in `weight` (300 rows) and `market_index` (374 rows): left as NaN — both XGBoost and CatBoost natively handle missing values by learning an optimal split direction for them, so no imputation was required.
- **Outlier audit across all numeric columns** using the IQR method, followed by manual verification of every flagged column. `Longitude` and `quote_signal` were flagged by IQR but confirmed to be naturally skewed, valid distributions (not errors) upon inspection. `Distance` and `posted_rate` outliers were confirmed to be legitimate long-haul / high-value loads, not data errors.
- **Target skew**: `posted_rate` is right-skewed — 255 loads (~0.5%) exceed \$6,457 (the IQR upper bound), some as high as \$17,900. These are longer-haul, disproportionately Reefer/Flatbed loads with a rate-per-mile roughly 3x the norm — a legitimate rare market segment, not noise. The model was trained on $\log_{1p}(\text{posted_rate})$ to reduce the influence of this skew on the loss function, with predictions inverse-transformed (`expm1`) before every evaluation and output step.

Feature engineering

`date` was decomposed into `day_of_week`, `month`, `day_of_year`, and `year` to expose seasonal and weekly patterns to the model (freight demand varies by both season and day of week). `pickup`, `delivery`, and `equipment` were treated as native categorical features — pandas `'category'` dtype for XGBoost, `string` dtype with CatBoost's built-in ordered target statistics for CatBoost — rather than one-hot encoded, since both models handle categorical splits natively and this avoids the dimensionality blow-up from one-hot encoding high-cardinality city columns.

Model comparison on the held-out test set

Model	MAE (\$)	RMSE (\$)	R-squared	MAPE
XGBoost	125.51	539.71	0.8638	5.84%
CatBoost (selected)	89.53	526.51	0.8704	4.21%

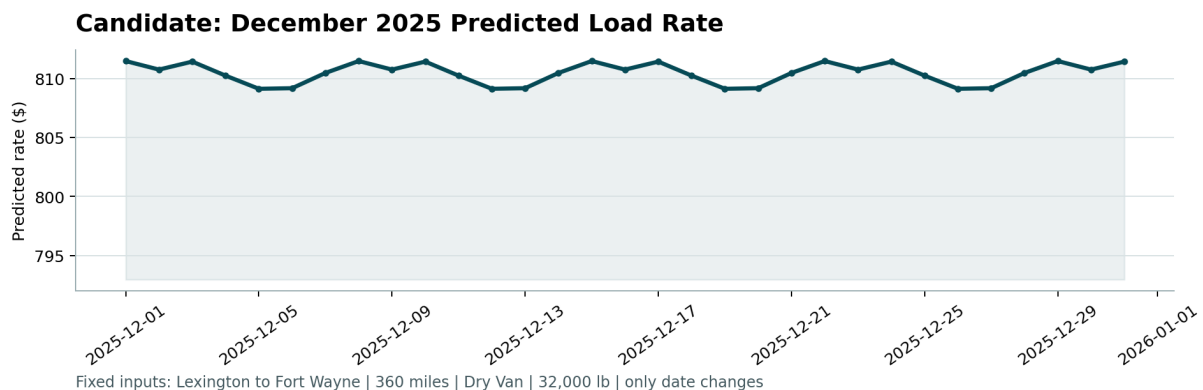
CatBoost was selected as the final model (lower MAE, RMSE, MAPE and higher R-squared across the board). The gap is largely attributable to CatBoost's ordered target-statistics encoding, which handles the high-cardinality pickup/delivery city columns more effectively than XGBoost's native categorical splits. A residual analysis confirmed both models' largest errors overlap heavily (19/20 of the worst-predicted test rows were shared between the two models) and correspond almost entirely to the rare, high-value load segment (`posted_rate > $6,457`) described above — a data characteristic rather than a model-specific weakness.

Validation-set inference

The selected CatBoost model — trained once on `train-test.csv` — was applied via `.predict()` to all 12,000 rows of `validation.csv` to produce `validation_predictions.csv` (`load_id`, `predicted_rate`). 1,447 of those rows (12.1%) reference pickup/delivery cities not present in the training data; these are flagged internally as lower-confidence, since CatBoost falls back to feature-based estimation rather than a learned city-specific effect for unseen categories.

2. Fixed December 2025 Prediction Chart

The same trained CatBoost model (no retraining) was used to predict rates for a fixed load — Lexington to Fort Wayne, Dry Van, 360 miles, 32,000 lb — across every day of December 2025, with only the date changing. Route-specific proxy values (median `market_index` and `quote_signal`, and exact pickup/delivery coordinates) were pulled from the 21 historical Lexington→Fort Wayne Dry Van loads in the training data, since the December template does not include these columns. This chart was produced by the provided `scorer.py` script from `data/december_chart_inputs.csv` after the `predicted_rate` column was filled in.



The predicted rate follows a clear 7-day cycle — modestly higher on weekdays and lower around weekends — consistent with typical freight demand patterns, indicating the model has learned a sensible day-of-week effect rather than an arbitrary one.